Shaan Patel

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Objective

Ambitious third-year Computer Engineering student specializing in computing hardware, VLSI design, and emerging architectures. Seeking a Fall 2026 or Spring 2027 internship in computer architecture, embedded systems, or hardware design to apply strong programming skills and an end-to-end approach to creating complex systems.

Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science; GPA: 3.59 — *August 2023 – May 2027*
Major in Computer Engineering, Minor in Applications of Artificial Intelligence and Machine Learning

- Relevant Coursework: Data Structures and Algorithms, Embedded Systems Design, Advanced Microarchitecture Design
- Leadership: Captain of GT Ramblin' Raas, GTXR Executive Board, SiliconJackets Digital Design Team Lead

Technical Skills

Programming: C++, Verilog, C, VHDL, RISC-V, SystemVerilog, OpenGL, Python, Java, JavaScript, HTML/CSS, SQL, LaTeX
Platforms: Linux (Ubuntu, Slackware, Debian), Red Hat (OpenStack, CloudForms), OpenVMS, Solaris, Windows, MacOS
Hardware: FPGA, ESP32, IC Breadboard Circuits, Oscilloscope, Arduino, Raspberry Pi, ARM MBED, LaunchPad, 3D Printing
Frameworks: React, SFML, CUDA, PyTorch, AWS (S3, DynamoDB, EC2), Express, MongoDB, Pandas, Node.js, Kubernetes
Software: KiCAD, Altium, Cadence, Intel Altera Quartus II, LTspice, NI LabVIEW, gRPC, TCP/UDP Tunnels, Blender, Autodesk Fusion 360/AutoCAD, SolidWorks, Onshape, Visual Studio Code, GitHub, Wireshark, Adobe Creative Suite, Inkscape, Slack
Communication: Design proposals, technical reports, instruction manuals, presentations, email correspondence, training materials

Experience

AeroCore-V - Hardware-Accelerated SoC for Autonomous UAVs — Atlanta, GA
SiliconJackets Project Lead — *August 2025 – Present*

- Designed a RISC-V SoC in SystemVerilog with custom ALU extensions for single-cycle PID control math, tailored for low-latency flight adjustments on a DE10 validated with Intel Quartus Prime
- Implemented a two-level cache system with an LRU replacement policy optimized for high-frequency I2C/SPI GPS and IMU data streams from an ESP32 to reduce memory stalls by 62% compared to a single-level cache design
- Developed a virtual memory system with a protected frame table to isolate flight-critical control routines from telemetry code, ensuring system resilience against page faults
- Orchestrated a priority-based thread scheduler to manage UAV flight loops and telemetry tasks, achieving a 15% latency reduction validated via a networked OpenGL Digital Twin physics simulation

Georgia Tech HIVE Makerspace — Atlanta, GA
Student Leader and Volunteer — *August 2024 – Present*

- Instruct ∼40 students per week on advanced design and fabrication workflows, including 3D CAD/slicing, circuit analysis with oscilloscopes, laser cutter operation, PCB design and manufacturing (taking a blank FR4 sheet to an operational PCB)
- Deployed a multi-device print manager to automate jobs on a FIFO basis, increasing the operational uptime of the free-to-use 12-unit Bambu Lab 3D print farm for the Georgia Tech student body
- Facilitate workshops and created training resources that contributed to a 20% increase in student participation
- Architecting a full-stack inventory management system using an AWS RDS backend to track materials in the cloud with S3 and DynamoDB to enable a student electronics rental program, aiming to reduce annual student costs

High-Performance Heterogeneous Computing Infrastructure — Ashburn, VA
BlendFarm Project Leader — *June 2022 – July 2025*

- Assembled two 2.5kW parallel computing systems from the component level, integrating twelve 12GB GDDR6 GPUs and modifying voltage curves to bypass LHR limiters, achieving a 73% hash rate improvement
- Overcame thermal throttling by designing and CNC-fabricating monolithic aluminum water blocks with Solidworks to cool a 6-GPU array, reducing core temperatures by over 15°C under sustained 100% load
- Scripted a distributed render farm using C++ and SFML to coordinate large ray tracing frame rendering across a network
- Optimized task allocation using TCP/UDP socket tunnels and a custom jitter buffer implementation to ensure reliable data synchronization between the master node and rendering clients for faster computer graphics generation

Academy of Science International Research — Ashburn, VA
Research Collaborator — *November 2022 – July 2024*

- Synthesized a saline-soluble polymer as a novel alternative to currently used petrochemical-based plastics as a means of preventing marine pollution
- Modeled and fabricated 3D-generated molds using filament and resin printers for synthesizing hydrogels through incubation
- Worked with a tensile tester to measure Young's modulus and SEM electron microscopy to analyze sample composition
- Led collaboration efforts with South Korean research teams for two years to formulate samples and complete testing
- Collected over 150 trials of data and was awarded the Yale Outstanding Project award at RSEF in Spring of 2023